

Interpolation

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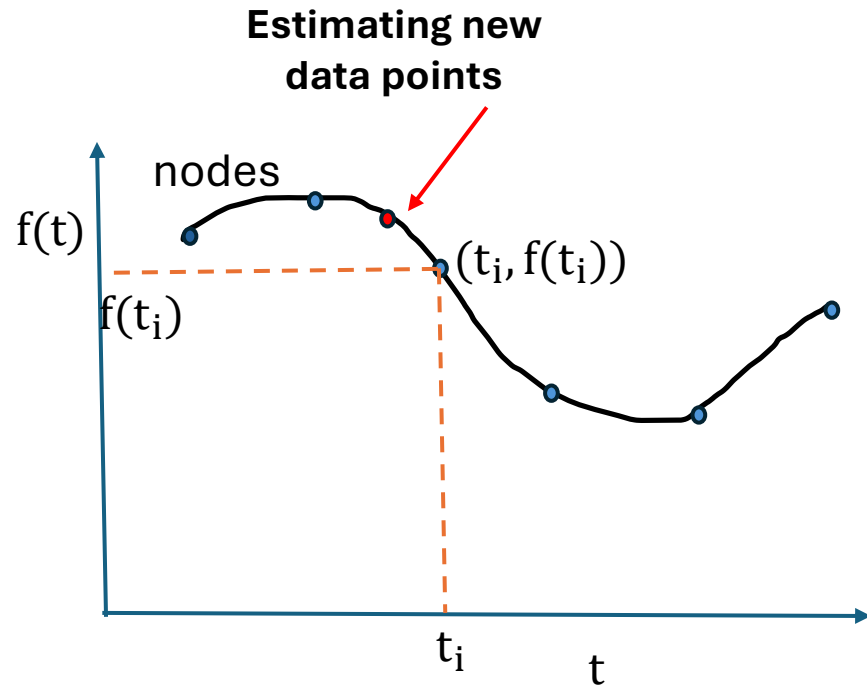
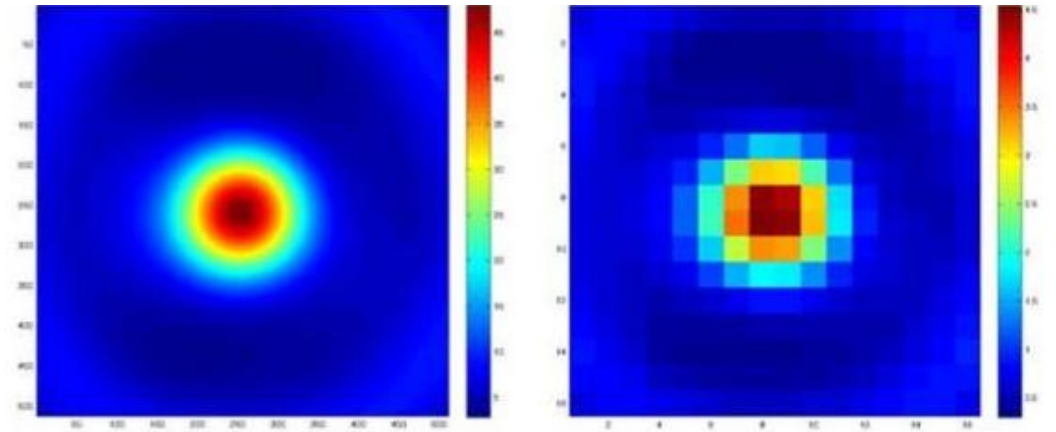


Image smoothing



Lagrange Interpolation:

Given a set of coordinate pairs $(x_i, f(x_i))$, $f(x)$ can be interpolated by a function $L(x)$, such that

$$L(x_i) = f(x_i)$$

Given a set of $k+1$ nodes, the interpolating function $L(x)$ is spanned by a basis of k^{th} order polynomial functions $l_j(x)$, i. e.,

$$L(x) = \sum_{i=1}^{k+1} f(x_i) l_i(x)$$

$$l_i(x) = \frac{(x - x_1)(x - x_2) \dots (x - x_{k+1})}{(x_i - x_1)(x_i - x_2) \dots (x_i - x_{k+1})} \quad (\text{Lagrange polynomial})$$

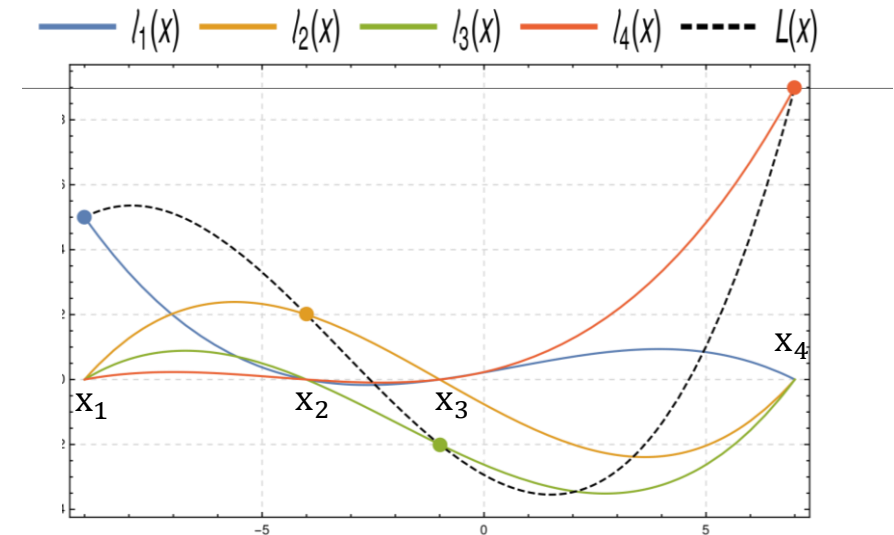
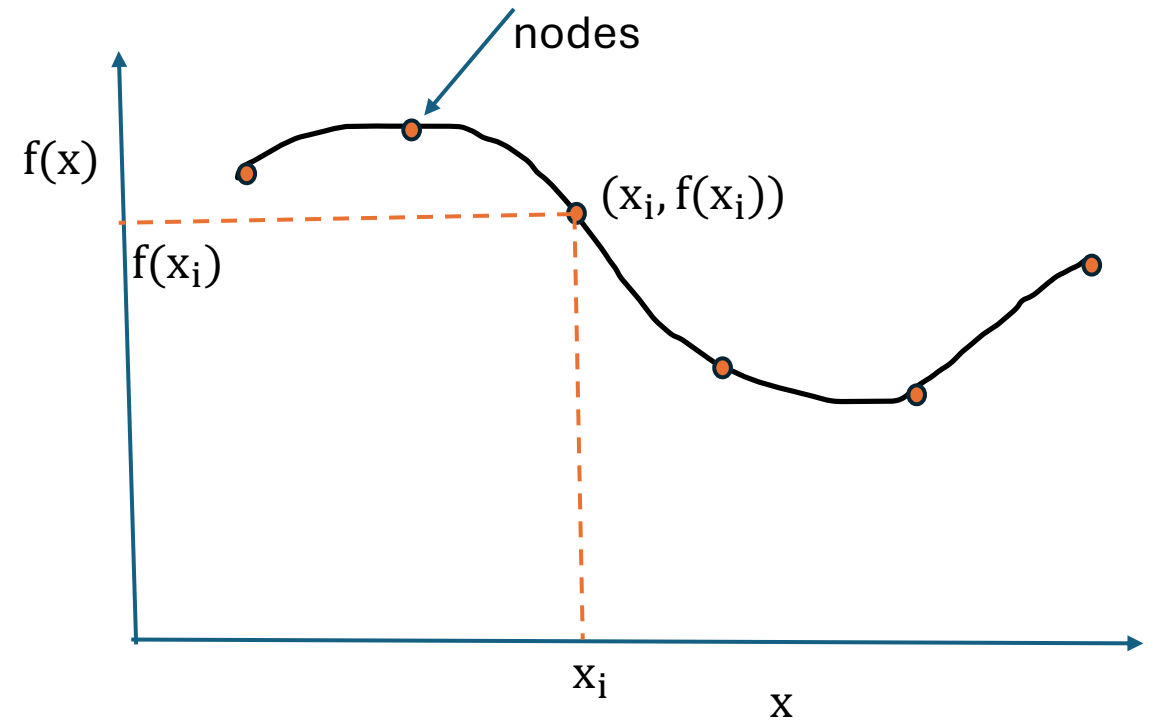
$$l_i(x) = \prod_{m=1, m \neq i}^{k+1} \frac{(x - x_m)}{(x_i - x_m)}$$

$$l_i(x_j) = \delta_{ij}$$

(Property of the Lagrange basis functions)

Polynomial interpolating 4 nodes:

$$L(x) = f(x_1) \frac{(x - x_2)(x - x_3)(x - x_4)}{(x_1 - x_2)(x_1 - x_3)(x_1 - x_4)} + f(x_2) \frac{(x - x_1)(x - x_3)(x - x_4)}{(x_2 - x_1)(x_2 - x_3)(x_2 - x_4)} + \dots$$



THEOREM 3.2 Main Theorem of Polynomial Interpolation. Let $(x_1, y_1), \dots, (x_n, y_n)$ be n points in the plane with distinct x_i . Then there exists one and only one polynomial P of degree $n - 1$ or less that satisfies $P(x_i) = y_i$ for $i = 1, \dots, n$. ■

Prove by contradiction:

Assume there are two distinct polynomials, $P(x)$ and $Q(x)$ of degree $n-1$ that interpolate the same n points.

$$\longrightarrow P(x_1) = Q(x_1) = y_1, P(x_2) = Q(x_2) = y_2, \dots, P(x_n) = Q(x_n) = y_n.$$

Define $H(x) = P(x) - Q(x)$.

$$\longrightarrow 0 = H(x_1) = H(x_2) = \dots = H(x_n);$$

$\longrightarrow H(x)$ is a $n-1$ degree polynomial with n roots!

$\longrightarrow H(x)$ is an identically-zero polynomial, i.e., $H(x)=0$

$$\longrightarrow P(x)=Q(x)$$

Representation of polynomials:

$$P(x) = a_0 + a_1x + a_2x^2 + \dots + a_{n-1}x^{n-1} = c_0 + c_1(x - x_1) + c_2(x - x_1)(x - x_2) + \dots + c_{n-1}(x - x_1)\cdots(x - x_{n-1}).$$

Choose $c_{n-1} = a_{n-1}$. Remaining $c_{n-2}, c_{n-3}, \dots, c_0$ recursively calculated by setting c_k to the coefficient of the k degree term.

Example: Find c_{n-2} as follows.

$$P(x) - c_{n-1}(x - x_1)\dots(x - x_{n-1}) = c_0 + \dots + c_{n-2}(x - x_1)\dots(x - x_{n-1})$$

Newton's Divided Difference:

Data points: $(x_1, f(x_1)), \dots, (x_n, f(x_n))$.

$$\begin{aligned} P(x) = & f[x_1] + f[x_1 \ x_2](x - x_1) \\ & + f[x_1 \ x_2 \ x_3](x - x_1)(x - x_2) \\ & + f[x_1 \ x_2 \ x_3 \ x_4](x - x_1)(x - x_2)(x - x_3) \\ & + \dots \\ & + f[x_1 \ \dots \ x_n](x - x_1) \dots (x - x_{n-1}). \end{aligned}$$

Where, $f[x_1 \dots x_n]$ the coefficient of the x^{n-1} term

$$\begin{aligned} f[x_k] &= f(x_k) \\ f[x_k \ x_{k+1}] &= \frac{f[x_{k+1}] - f[x_k]}{x_{k+1} - x_k} \\ f[x_k \ x_{k+1} \ x_{k+2}] &= \frac{f[x_{k+1} \ x_{k+2}] - f[x_k \ x_{k+1}]}{x_{k+2} - x_k} \\ f[x_k \ x_{k+1} \ x_{k+2} \ x_{k+3}] &= \frac{f[x_{k+1} \ x_{k+2} \ x_{k+3}] - f[x_k \ x_{k+1} \ x_{k+2}]}{x_{k+3} - x_k}, \end{aligned}$$

The recursive definition of the Newton's divided differences allows arrangement into a convenient table. For three points the table has the form

x_1	$f[x_1]$		
		$f[x_1 \ x_2]$	
x_2	$f[x_2]$		$f[x_1 \ x_2 \ x_3]$
		$f[x_2 \ x_3]$	
x_3	$f[x_3]$		

Example:

Use divided differences to find the interpolating polynomial passing through the points $(0, 1), (2, 2), (3, 4)$.

0	1		
		$\frac{1}{2}$	
2	2		$\frac{1}{2}$
		2	
3	4		

$$P(x) = 1 + \frac{1}{2}(x - 0) + \frac{1}{2}(x - 0)(x - 2),$$

Proof of Newton's interpolating polynomial:

The interpolating polynomial of $x_2, x_3, \dots, x_{k-1}, x_1, x_k$ is

$$P_1(x) = f[x_2] + f[x_2 x_3](x - x_2) + \dots + f[x_2 x_3 \dots x_{k-1} x_1](x - x_2) \cdots (x - x_{k-1}) \\ + f[x_2 x_3 \dots x_{k-1} x_1 x_k](x - x_2) \cdots (x - x_{k-1})(x - x_1)$$

and the interpolating polynomial of $x_2, x_3, \dots, x_{k-1}, x_k, x_1$ is

$$P_2(x) = f[x_2] + f[x_2 x_3](x - x_2) + \dots + f[x_2 x_3 \dots x_{k-1} x_k](x - x_2) \cdots (x - x_{k-1}) \\ + f[x_2 x_3 \dots x_{k-1} x_k x_1](x - x_2) \cdots (x - x_{k-1})(x - x_k).$$

$$P_1(x) = P_2(x) \text{ [by uniqueness property]}$$

$$P_1(x_k) = P_2(x_k) \implies f[x_2] + f[x_2 x_3](x_k - x_2) + \dots + f[x_2 x_3 \dots x_{k-1} x_1 x_k](x_k - x_2) \cdots (x_k - x_{k-1})(x_k - x_1) = \\ f[x_2] + f[x_2 x_3](x_k - x_2) + \dots + f[x_2 x_3 \dots x_{k-1} x_k x_1](x_k - x_2) \cdots (x_k - x_{k-1})(x_k - x_k)$$

$$\implies f[x_2 \dots x_{k-1} x_1](x_k - x_2) \cdots (x_k - x_{k-1}) + f[x_2 \dots x_{k-1} x_1 x_k](x_k - x_2) \cdots (x_k - x_{k-1})(x_k - x_1) \\ = f[x_2 \dots x_k](x_k - x_2) \cdots (x_k - x_{k-1})$$

$$\implies f[x_2 \dots x_{k-1} x_1] + f[x_2 \dots x_{k-1} x_1 x_k](x_k - x_1) = f[x_2 \dots x_k].$$

$$\implies f[x_1 \dots x_k] = \frac{f[x_2 \dots x_k] - f[x_1 \dots x_{k-1}]}{x_k - x_1}.$$

Interpolation Error:

Consider an $n-1$ degree polynomial $P_{n-1}(t)$ interpolating n data points.

Add a point x to the data points such that the new polynomial is

$$P_n(t) = P_{n-1}(t) + f[x_1 \dots x_n x](t - x_1) \cdots (t - x_n).$$

Evaluated at the extra point x , $P_n(x) = f(x)$, so

$$f(x) = P_{n-1}(x) + f[x_1 \dots x_n x](x - x_1) \cdots (x - x_n).$$

This formula is true for all x . Now define

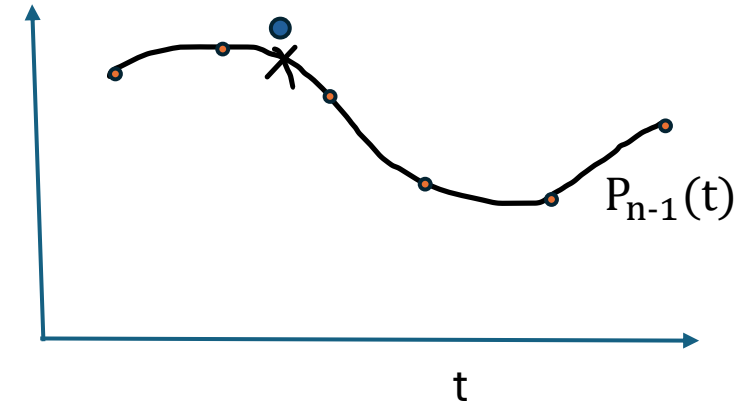
$$h(t) = f(t) - P_{n-1}(t) - f[x_1 \dots x_n x](t - x_1) \cdots (t - x_n).$$

Note that $h(x) = 0$, and $h(x_1) = h(x_2) = \dots h(x_n) = 0$

Between neighboring pairs of the $n + 1$ points $x, x_1, \dots, x_n$ there must be a point where $h' = 0$ by **Rolle's** theorem.

Similarly, between neighboring pairs of the subsequent n points there must be a point where $h'' = 0$.

Continuing this way, there must be a point c where $h^n(c) = 0$.



$$h^{(n)}(t) = f^{(n)}(t) - n!f[x_1 \dots x_n x],$$

because the n th derivative of the polynomial $P_{n-1}(t)$ is zero. Substituting c gives

$$f[x_1 \dots x_n x] = \frac{f^{(n)}(c)}{n!},$$

which leads to

$$f(x) = P_{n-1}(x) + \frac{f^{(n)}(c)}{n!}(x - x_1) \cdots (x - x_n),$$